# Pulley 1 Shaft

G

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$P = 50kW$

$P = T\omega$

$T_1 = \frac{P}{\omega}$

$= \frac{50 \times 10^3}{950 \times \frac{2\pi}{60}}$

$= 502.595 Nm$

$\tau_{allowable} = 40MPa$

$\tau_{allowable} = \frac{T}{\frac{\pi d^{\cancel{4}3}}{32}}\left(\frac{\cancel{d}}{2}\right)$

$\frac{502.595}{\frac{\pi d^3}{16}} \leq 40 \times 10^9$

$502.595 \leq 7.85 \times 10^9 d^3$

$d^3 \geq 63.99 \times 10^{-9}$

$d \geq 4.0 \times 10^{-3}$

$d \geq 40 mm$

Shaft diameter = 40 mm

## Determine design power

For a motor torque of 50kW and a service time of 12 hours

From table A-2:

Service factor = 1.2

$$\text{Design power} = 1.2 \times 50$$
$$= 60\,kW$$

## Belt cross section

From table A-3:

Belt cross section is SPB.

## Select sheave diameter

Assume belt speed to be $21 ms^{-1}$.

$$v_b = 21 ms^{-1}$$

$$v_b = r_1 w = \frac{D_1}{2} w$$

$$21 = \frac{D_1}{2}\left(2\pi\left(\frac{950}{60}\right)\right)$$

$$\frac{21}{\frac{950\pi}{60}} = D_1$$

$$D_1 = 0.422 m$$

From table A-4:

$$D_1 = 0.400 m$$

Checking the belt speed:

$$v_b = \frac{D_1}{2} w$$

$$= \frac{0.400}{2}\left(2\pi\left(\frac{950}{60}\right)\right)$$

$$= 19.9 ms^{-1}$$

Since $5 ms^{-1} < v_b < 33 ms^{-1}$, $D_1$ of $0.400 m$ is fine.

# Finding the other diameter

$$SR = \frac{n_1}{n_2} = \frac{D_2}{D_1}$$

$$n_2 = \frac{410+430}{2} = 420 \text{ rpm}$$

$$SR = \frac{950}{420} = 2.262$$

The other sheave diameter:

$$D_2 = D_1 (2.262)$$
$$= 0.4(2.262)$$
$$= 0.905 \text{ m}$$

From table A-4:

$$D_2 = 0.900 \text{ m}$$

Check if $410 \leq n_2 \leq 430$

$$n_2 = \frac{0.4}{0.9}(950)$$
$$= 422 \text{ rpm}$$

(within range?)

Since $410 \leq n_2 \leq 430$, it is within range.

Actual speed ratio:

$$SR = \frac{n_1}{n_2} = \frac{0.9}{0.4} = 2.25$$

Tentative centre distance = 1.4 m

Tentative belt length (TBL):

$$TBL = 2 \times TCD + 1.57(D_2 + D_1) + \frac{(D_2 - D_1)^2}{4TCD}$$

$$= 2 \times 1.4 + 1.57(0.9 + 0.4) + \frac{(0.9 - 0.4)^2}{4(1.4)}$$

$$= 4.886 \text{ m}$$

From table A-1:

Use belt length of 5 m for convenience.

$$\therefore L = 5 \text{ m}$$

Actual centre distance:

$$C = \frac{B + \sqrt{B^2 - 32(D_2 - D_1)^2}}{16}$$

$$= \frac{4L - 6.28(D_2 - D_1) + \sqrt{(4L - 6.28(D_2 - D_1)) - 32(D_2 - D_1)^2}}{16}$$

$$= 1.458 \text{ m}$$

# Power Correction Factor

Angle of contact on the smaller sheave:

$$\theta_1 = 180° - 2\arcsin\left(\frac{D_2 - D_1}{2C}\right)$$

$$= 180 - 2\arcsin\left(\frac{0.9 - 0.4}{2(1.458)}\right)$$

$$= 160.25°$$

From table A-5:

$$\frac{160.25 - 157}{163 - 157} = \frac{C_\theta - 0.94}{0.96 - 0.94}$$

$$C_\theta \simeq 0.95$$

From table A-6:

$$C_L = 1.06$$

## Determine rated power

From Table 7c:

$SR = 2.25$

$$\text{Rated power} = \frac{2.25-1.5}{3.0-1.5}(23.21-23.07)+23.07$$

$$= 23.14 \text{ kW belt}^{-1}$$

## Finding the corrected rated power (CRP)

$$CRP = C_\theta C_L (RP)$$

$$= 0.95(1.06)(23.14)$$

$$= 23.3 \text{ kW belt}^{-1}$$

Number of belts:

$$n = \frac{\text{Design power}}{CRP} = \frac{60}{23.3}$$

$$= 2.57$$

$$\approx 3$$

# Pulley 2 Shaft

$$P = 50\,kW$$

$$n_2 = 422\,rpm$$

$$T_2 = \frac{P}{\omega_2}$$

$$= \frac{50\times10^3}{422\left(\frac{2\pi}{60}\right)}$$

$$= 1131.43\,Nm$$

$$\tau_{allowable} = 40\,mPa$$

$$\tau_{allowable} = \frac{T}{\frac{\pi d^{\cancel{4}3}}{\cancel{32}\,16}}\left(\frac{d}{2}\right)$$

$$40\times10^9 \geq \frac{1131.43}{\frac{\pi d^3}{16}}$$

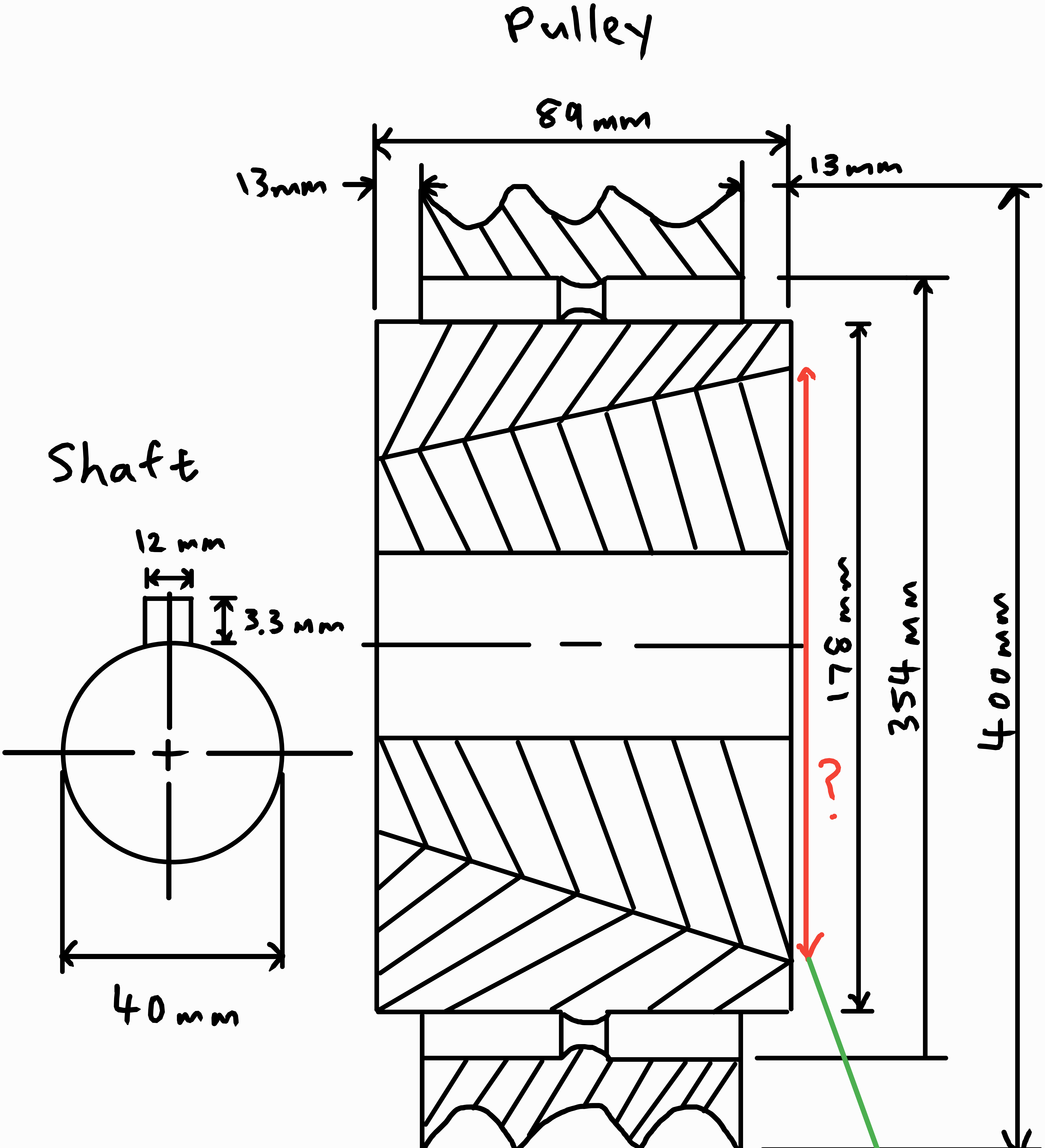
$$d^3 \geq 144.058\times10^{-9}$$

$$d \geq 5.242\times10^{-3}\,m$$

$$\geq 52.4\,mm$$

$$\therefore d_2 = 55\,mm$$

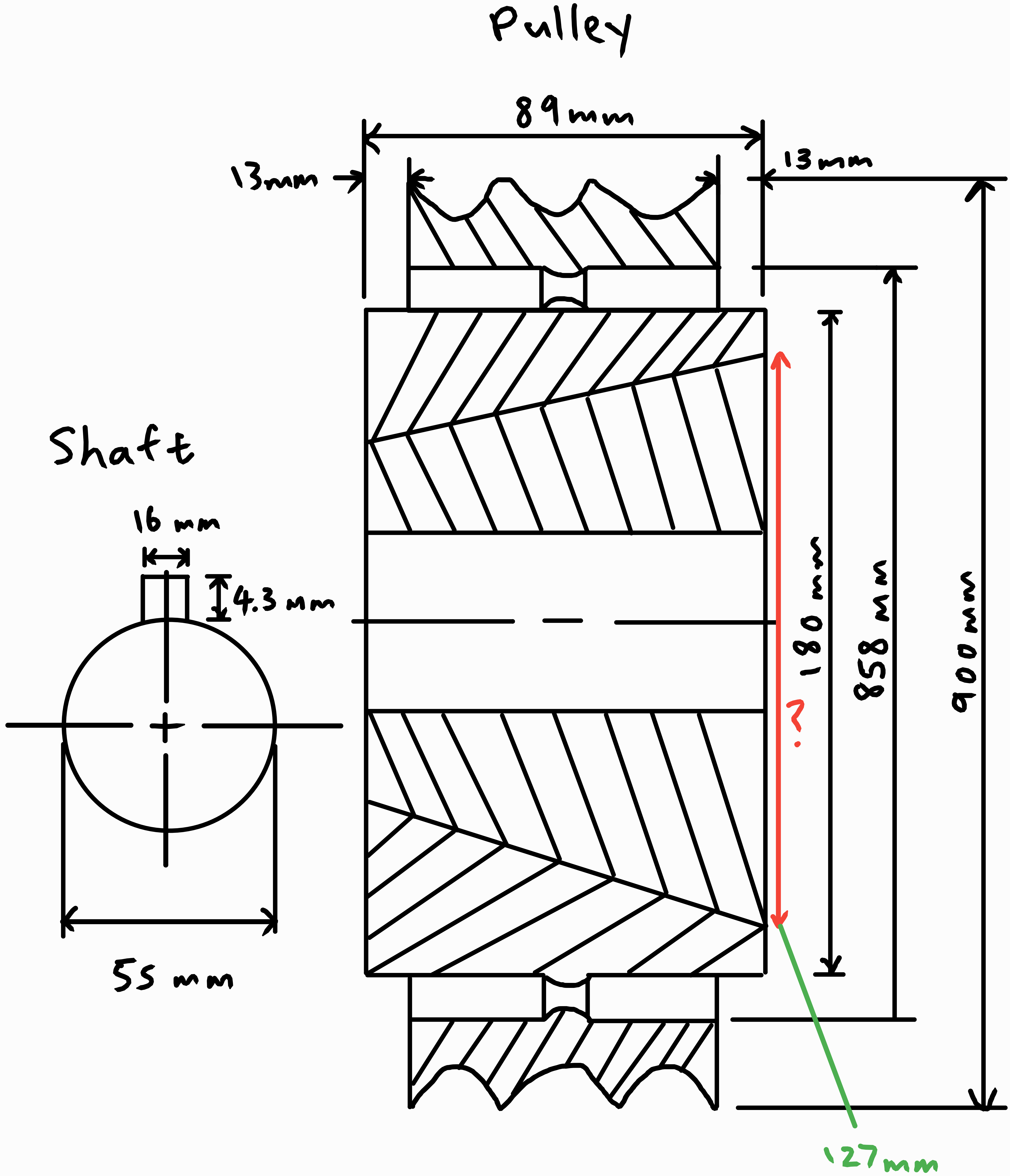
# Pulley 1




Bush number? 3535 127mm

Look at the catalogue for the taper lock pulleys to find a suitable bush.

# Pulley 2

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